Predictive Trading Strategy

Standalone signal engine for crypto / equities backtesting

1. Overview

The Predictive Strategy is a rule-based trading signal engine that combines multiple technical indicators into a scoring system. It generates buy, sell, or hold signals with associated stop-loss and take-profit levels. The strategy is designed for backtesting on historical OHLCV (Open, High, Low, Close, Volume) candle data.

2. Data Model

Each price candle is represented as a Candle dataclass with the following fields:

- ts - Timestamp (integer)
- o - Open price
- h - High price
- l - Low price
- c - Close price
- v - Volume

3. Technical Indicators

3.1 EMA (Exponential Moving Average)

Calculated with smoothing factor k = 2 / (period + 1). The first value is the simple mean of the first 'period' prices; subsequent values are computed as:

EMA = price * k + previous_EMA * (1 - k)

Used as the building block for MACD.

3.2 RSI (Relative Strength Index)

Standard 14-period RSI. Computed by averaging gains and losses over the lookback window:

RS = avg_gain / avg_loss

RSI = 100 - (100 / (1 + RS))

Thresholds: oversold at 35, overbought at 65.

3.3 MACD (Moving Average Convergence Divergence)

Built from three EMAs:

MACD Line = EMA(close, 12) - EMA(close, 26)

Signal = EMA(MACD Line, 9)

Histogram = MACD Line - Signal

Trading signals are derived from the histogram direction relative to the signal line.

3.4 SMA (Simple Moving Average)

50-period simple moving average used as a trend filter. Price above MA50 suggests uptrend; price below suggests downtrend.

4. Signal Generation Logic

4.1 RSI / Price Divergence

Divergence is detected by sampling three price lows and RSI values across the last 10 candles (every 3rd candle):

Bullish divergence: Price makes lower low, RSI makes higher low.

Bearish divergence: Price makes higher low, RSI makes lower low.

Divergence is the strongest signal and contributes 3 points to the scoring system.

4.2 MACD Signal Prediction

The algorithm checks the histogram slope over a 2-bar lookback:

MACD buy = histogram rising AND MACD line below signal line.

MACD sell = histogram falling AND MACD line above signal line.

MACD signals contribute 2 points.

4.3 Trend & Volume Filters

Each of the following conditions contributes 1 point to the respective score:

Trend: Price vs. 50-period SMA (above = bullish, below = bearish)

RSI extreme: Below 35 (oversold, bullish) / above 65 (overbought, bearish)

Volume spike: Current volume > 1.5x the 20-bar average volume

5. Scoring & Entry Rules

Buy and sell scores are accumulated independently. A signal is generated when:

1. The dominant score exceeds 3 (e.g., buy_score >= 3).
2. The dominant score is strictly greater than the opposite score.

If conditions are not met, the engine returns 'hold'.

Signal Types

- Buy: Direction = 'buy', reason = 'div' (divergence) or 'macd'
- Sell: Direction = 'sell', reason = 'div' or 'macd'
- Hold: No actionable signal; skip this bar

6. Risk Management (Stop-Loss & Take-Profit)

Every entry signal includes fixed percentage stop-loss and take-profit levels relative to the entry price:

Buy entry: SL = price * 0.94 (6% below entry)
TP = price * 1.12 (12% above entry)

Sell entry: SL = price * 1.06 (6% above entry)
TP = price * 0.90 (10% below entry)

The reward-to-risk ratio is 2:1 for buys and approximately 1.67:1 for sells.

7. Known Limitations

- No dynamic position sizing or capital management.
- Fixed SL/TP levels do not adapt to volatility (e.g., ATR-based).
- Divergence detection uses a simple 3-point heuristic, not a peak/trough algorithm.
- No trade filtering by broader market regime or trend strength (ADX).
- Backtest-only logic; no live data feed or order execution integration.

8. Usage

```
from predictive_strategy import Candle, generate_signal

candles = [Candle(ts=..., o=..., h=..., l=..., c=..., v=...), ...]
direction, reason, sl, tp = generate_signal(candles, idx=-1)
```

See predictive_strategy.py (132 lines) for the full implementation.